

Ibn Tofail University

Analysis IV

Problem Set III

Exercise 1:

Study the nature of convergence of each of the following sequences of functions:

1. $f_n(x) = \frac{1}{1+n(nx-1)^2}$, for $x \in I = [0, 1]$.

2. $f_n(x) = \begin{cases} (n-1)x & \text{if } x \in [0, \frac{1}{n}] , \\ 1-x & \text{if } x \in [\frac{1}{n}, 1] . \end{cases}$

3. $f_n(x) = e^{-nx} \cos x^2$, for $x > 0$.

Correction

1. Pointwise limit: $f(x) = 0$ on $[0, 1]$.
 Uniform convergence? $\sup_{x \in [0,1]} |f_n(x) - 0| = f_n(1/n) = 1 \not\rightarrow 0$.
 Convergence is **pointwise but not uniform**.
2. Pointwise limit: $f(x) = 1 - x$ on $[0, 1]$.
 Uniform norm: $\sup_{x \in [0,1]} |f_n(x) - f(x)| = \sup_{x \in [0,1/n]} |nx - 1| = 1 \not\rightarrow 0$.
 Convergence is **pointwise but not uniform**.
3. Pointwise limit: $f(x) = 0$ on $(0, +\infty)$.
 Uniform norm: consider $x_n = 1/n$, then $|f_n(x_n)| = e^{-1} \cos(1/n^2) \rightarrow e^{-1} \neq 0$.
 Convergence is **pointwise but not uniform** on $(0, +\infty)$.

Exercise 2:

Consider the sequence of functions $f_n : [-1, 1] \rightarrow \mathbb{R}$, defined by

$$x \mapsto f_n(x) = e^{-nx^2} \sin nx + \sqrt{1-x^2}, \quad n \in \mathbb{N}.$$

1. Show that the sequence of functions (f_n) converges pointwise on $I = [-1, 1]$ to a function f , which you will determine.
2. Show that (f_n) converges uniformly to f on $[\omega, 1]$, where ω is a positive constant.
3. Show that (f_n) does not converge uniformly to f on $[0, 1]$.

Correction

- For $x = 0$: $f_n(0) = 1 \rightarrow 1$.
For $x \neq 0$: $e^{-nx^2} \sin nx \rightarrow 0$, so $f_n(x) \rightarrow \sqrt{1-x^2}$.
Pointwise limit: $f(x) = \begin{cases} 1 & x = 0 \\ \sqrt{1-x^2} & x \in [-1, 1] \setminus \{0\} \end{cases}$.
- For $x \in [\omega, 1]$, $|f_n(x) - f(x)| = |e^{-nx^2} \sin nx| \leq e^{-n\omega^2} \rightarrow 0$.
Convergence is **uniform** on $[\omega, 1]$.
- Choose $x_n = \pi/(2n)$. Then $|f_n(x_n) - f(x_n)| = e^{-\pi^2/(4n)} \rightarrow 1 \neq 0$.
Convergence is **not uniform** on $[0, 1]$.

Exercise 3:

Determine the domain of convergence D of each series:

- $\sum_{n>0} \frac{\cos nx}{n!}$.
- $\sum_{n>1} \frac{1}{(z-n)^2}$.
- $\sum_{n>1} \left| \sin \frac{x}{2^n} - \tan \frac{x}{2^n} \right|^{\frac{1}{2}}$.
- $\sum_{n>1} \frac{n!(x-3)^{2n}}{n^{n+1}}$.

Correction

- $|u_n(x)| \leq 1/n!$, $\sum 1/n!$ converges.
 $D = \mathbb{R}$ (absolutely and uniformly convergent).
- For fixed z , $|u_n(z)| \sim 1/n^2$, so converges for $z \in \mathbb{C} \setminus \{2, 3, 4, \dots\}$.
 $D = \mathbb{C} \setminus \{n \in \mathbb{N} : n \geq 2\}$.
- As $n \rightarrow \infty$, $u_n(x) \sim \frac{|x|^{3/2}}{\sqrt{2}} \cdot 2^{-3n/2}$, geometric decay.
 $D = \mathbb{R}$.
- Ratio test: $L(x) = |x-3|^2/e$.
Converges if $|x-3| < \sqrt{e}$, diverges if $|x-3| > \sqrt{e}$.
 $D = (3 - \sqrt{e}, 3 + \sqrt{e})$.

Exercise 4:

Consider the following series of functions:

$$g_n(x) = \frac{\sin x}{x^2 + 1} + \frac{\sin 2x}{x^2 + 2^2} + \dots + \frac{\sin nx}{x^2 + n^2} + \dots$$

- Determine the domain of convergence D .
- Study the uniform convergence on D .

Correction

1. $|u_n(x)| \leq 1/n^2$, $\sum 1/n^2$ converges.
By Weierstrass M-test: $D = \mathbb{R}$.
2. Since $|u_n(x)| \leq 1/n^2$ and $\sum 1/n^2$ converges,
convergence is **uniform** on $\mathbb{R}$.

Exercise 5:

Consider the series of functions with general term:

$$f_n(x) = \frac{nx}{1+n^2x^2} - \frac{nx+x}{1+n^2x^2+2n^2x+x^2}$$

1. Determine the domain of convergence D .
2. Calculate the sum of the series.
3. What is the nature of convergence of this series on D ?
4. Determine the nature of convergence of this series on $I =]0, +\infty[$ and $I = [a, +\infty[$ with $a > 0$.

Correction

1. The series telescopes: $f_n(x) = u_n(x) - u_{n+1}(x)$ with $u_n(x) = \frac{nx}{1+(nx)^2}$.
Partial sum: $S_N(x) = u_1(x) - u_{N+1}(x) \rightarrow u_1(x)$.
 $D = \mathbb{R}$.
2. Sum: $S(x) = u_1(x) = \frac{x}{1+x^2}$.
3. Remainder: $|R_N(x)| = \left| \frac{(N+1)x}{1+((N+1)x)^2} \right|$.
 $\sup_{x \in \mathbb{R}} |R_N(x)| = 1/2 \not\rightarrow 0 \Rightarrow$ **not uniform** on $\mathbb{R}$.
4. On $]0, +\infty[$: still not uniform (supremum $1/2$).
On $[a, +\infty[$ with $a > 0$: $\sup_{x \geq a} |R_N(x)| = \frac{(N+1)a}{1+((N+1)a)^2} \rightarrow 0 \Rightarrow$ **uniform**.

Exercise 6:

Consider the series of functions with general term: $f_n(x) = (\sin x)^\alpha (\cos x)^n$.

1. Determine the sum and domain of convergence of this series.
2. Show that we have uniform convergence on any compact $[a, b]$ included in $]0, \frac{\pi}{2}[$.

Correction

1. Geometric series: $\sum (\sin x)^\alpha (\cos x)^n = (\sin x)^\alpha \cdot \frac{1}{1 - \cos x}$
Converges if $|\cos x| < 1$, i.e., $x \notin \pi\mathbb{Z}$.
Domain: $D = \mathbb{R} \setminus \pi\mathbb{Z}$ (assuming α integer).
2. On $[a, b] \subset (0, \pi/2)$:
 $|f_n(x)| \leq (\sin b)^\alpha (\cos a)^n = M_n$.
 $\sum M_n$ converges (geometric with ratio $\cos a < 1$).
By Weierstrass M-test: **uniform** convergence.